

MONASH
University

Submission 2: Storyboarding and Low-Fidelity Prototypes

Student name:

Nguyen Khang Huynh - 33326460

Table of Contents

Table of Contents.....	1
1. Introduction.....	2
2. Selected User Stories.....	2
Original user stories.....	2
Modifying user stories.....	2
3. Storyboards.....	3
4. List of Acceptance Criteria for each user story.....	5
5. Low-Fidelity Prototypes.....	6
6. Design Report.....	8
Screen code: LACY001.....	8
Screen code: LACY002.....	9
Screen code: LACY003.....	10
Conclusion.....	11
Appendix.....	12
Relevant Personas (from submission 1, based on selected user stories).....	12

1. Introduction

This document aims to identify the requirements of the persona, Lacy Ho [1], and demonstrates design solutions to address these requirements. This document will bring back the user stories from the previous submission and make some necessary modifications. Based on the modified user stories, it will develop a set of storyboards to illustrate the scenario that reflect user interactions. The storyboard will provide a clear user requirement to develop a Kanban board containing a list of acceptance criteria which provide a functional features solution. The three-screen low-fidelity prototypes will be created following the acceptance criteria, along with Norman's Design Principles and Accessibility Principles. In the end, this document will analyze how these principles have been applied, outline the limitations of low-fidelity prototypes and two main goals for the high-fidelity prototypes.

2. Selected User Stories

Original user stories

1. As Lacy, I want to have many options of cuisine and desserts to learn so that I can have multiple types of food options to cook for my family meal. (should)
2. As Lacy, I want to know details of the ingredients such as price and places to buy so that I can consider the price and effort when finding it. (must)

Modifying user stories

1. *As Lacy, I want to easily find various cuisines and dishes to learn, so that I can have various meal options to cook for my family.*

Justification: The modification from "have" to "easily find" enhances that Lacy requires product support so that she can easily find and learn more cuisines and dishes to cook for her family and improve her cooking knowledge and skills.

2. *As Lacy, I want to know up-to-date details of the ingredients such as price and places to buy, so that I can find the appropriate ingredients with minimal effort.*

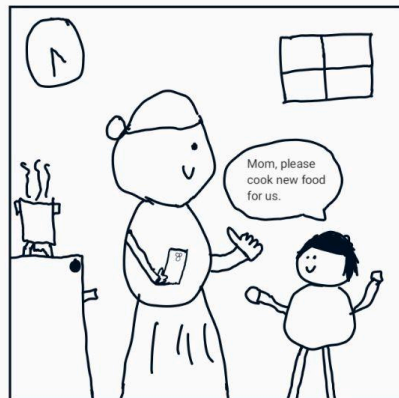
Justification: The modification by adding "up-to-date" and "minimal effort" enhances that Lacy wants to have newest and correct ingredients details and spend less resource and effort on achieving the goal.

3. Storyboards

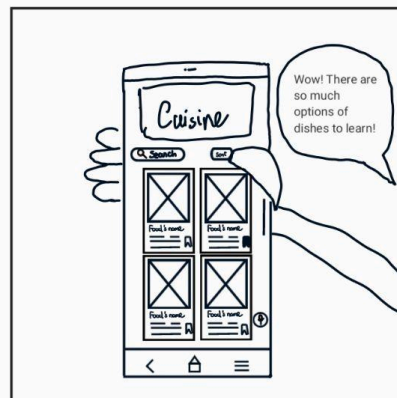
Scenario: Lacy's son wants Lacy to know more cuisines to cook for their dinner.

User story: *As Lacy, I want to easily find various cuisines and dishes to learn, so that I can have various meal options to cook for my family.*

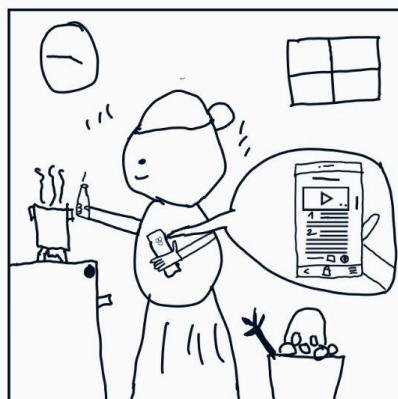
Accessibility: Situational, using voice command to automatically save the lesson whenever they can not manually save it.



Lacy's son want to try out new dishes/cuisines. Lacy is happy to learn new cuisines to cook for her family.



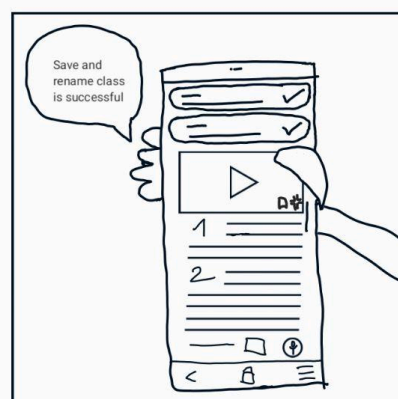
Lacy is suprised with various of options to learn. She decide to look through it and choose out one of them.



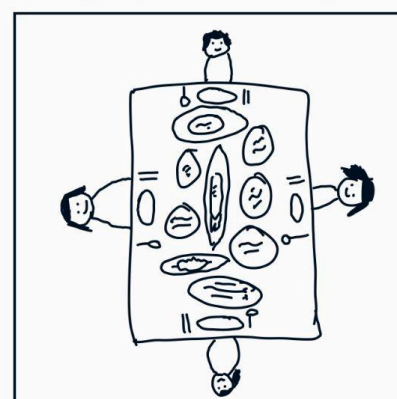
Videos and captions provide a structure and easy-to-follow instructions despite lacking of cooking skills. She easy to complete all of the steps.



Lacy want to save it and rename as "Dinner" for future use. However, her hand is not free so she decides to activate app's AI by her voice and status bar appear.



The app responds back to Lacy whenever it success with a voice announcement and text notification.



Lacy is excited because she manages to cook various of cuisines and dishes for her family. Everyone is happy and enjoys Lacy's food.

Photo 1: Storyboard 1

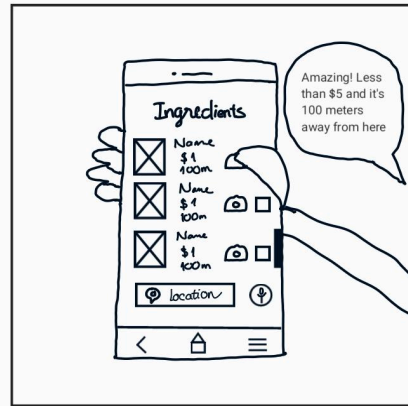
Scenario: Lacy wants to buy appropriate cheap ingredients with minimal effort.

User story: As Lacy, I want to know up-to-date details of the ingredients such as price and places to buy, so that I can find the appropriate ingredients with minimal effort.

Accessibility: Permanent, users with vision issues still know if they buy the appropriate items.



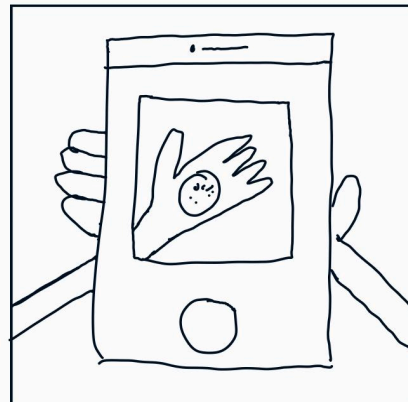
Lacy, a office worker, is preparing ingredients for a cooking attempt. She is on the apps looking at ingredients details.



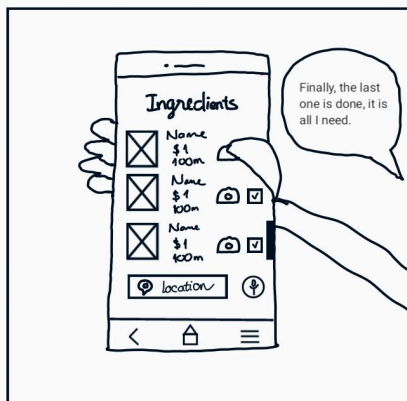
Lacy is excited about the cheap prices and the store is close to her place.



At the store, Lacy is excited because she can easily find those ingredients with a up-to-date prices.



Lacy takes the corresponding quantity and scan it through the app to validate because it quiet small for her to see it.



The app will tick the box whenever the ingredients is matched and the scanning process is completed.



Lacy arrives back home with all of the ingredients she need and she is suprised that it takes her only 30 minutes to buy groceries

Photo 2: Storyboard 2

4. List of Acceptance Criteria for each user story

User stories	To Do	Doing	Done
<i>As Lacy, I want to easily find various cuisines and dishes to learn, so that I can have various meal options to cook for my family.</i>			• a1. A list of clickable cards with text and image. • a2. A save button on each card saves the chosen class. • a3. A sort button to sort all options. • a4. A filter button to filter all options based on their category. • a5. A search bar to find any cuisine by name. • a6. Videos are accompanied by instructions in text format. • a7. A play button starts the video. • a8. A zoom button to expand the video. • a9. A caption button to turn on the video caption.
<i>As Lacy, I want to know up-to-date details of the ingredients such as price and places to buy, so that I can find the appropriate ingredients with minimal effort.</i>			• b1. A button to activate the ingredient page. • b2. A camera button activates ingredients scanning • b3. A grocery list displays items with photos and text descriptions. • b4. A location button shows the nearest store location • b5. A checkbox allows user marks found ingredients. • b6. A home button returns to the home page. • b7. A back button returns to the previous page. • b8. A menu button access to the app's menu • b9. A notification announces the success of any action. • b10. A top bar demonstrates the status of running AI tools. • b11. A button to activate the AI tool.

5. Low-Fidelity Prototypes

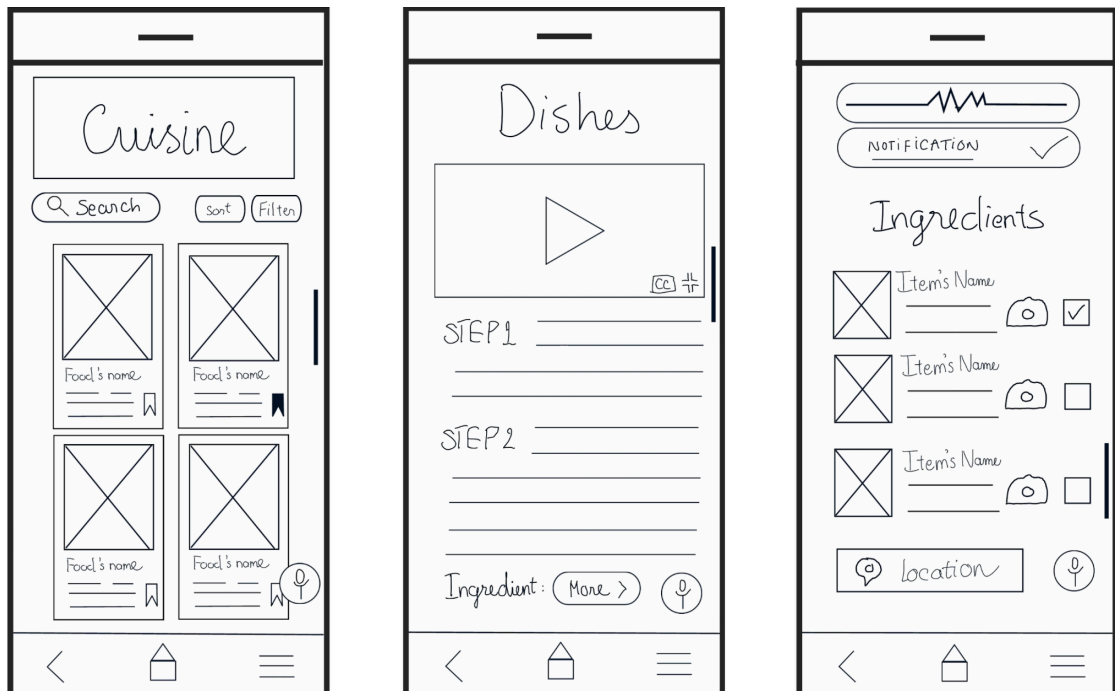


Photo 3: Three screens without annotation.

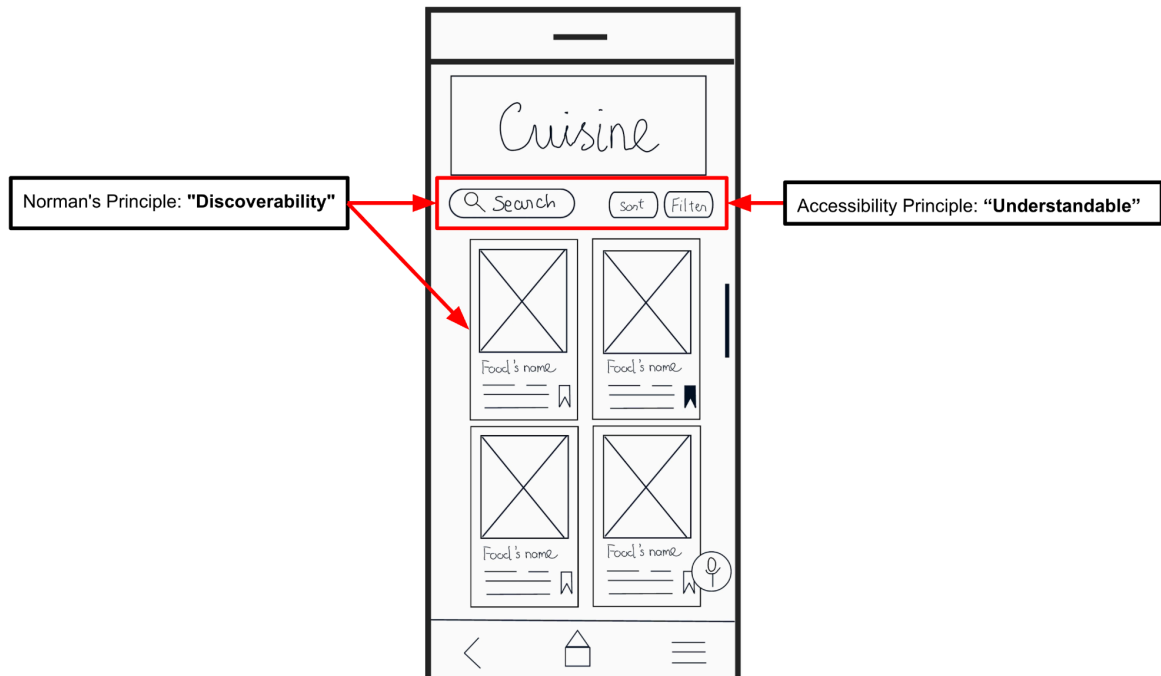


Photo 4: First screen with annotation

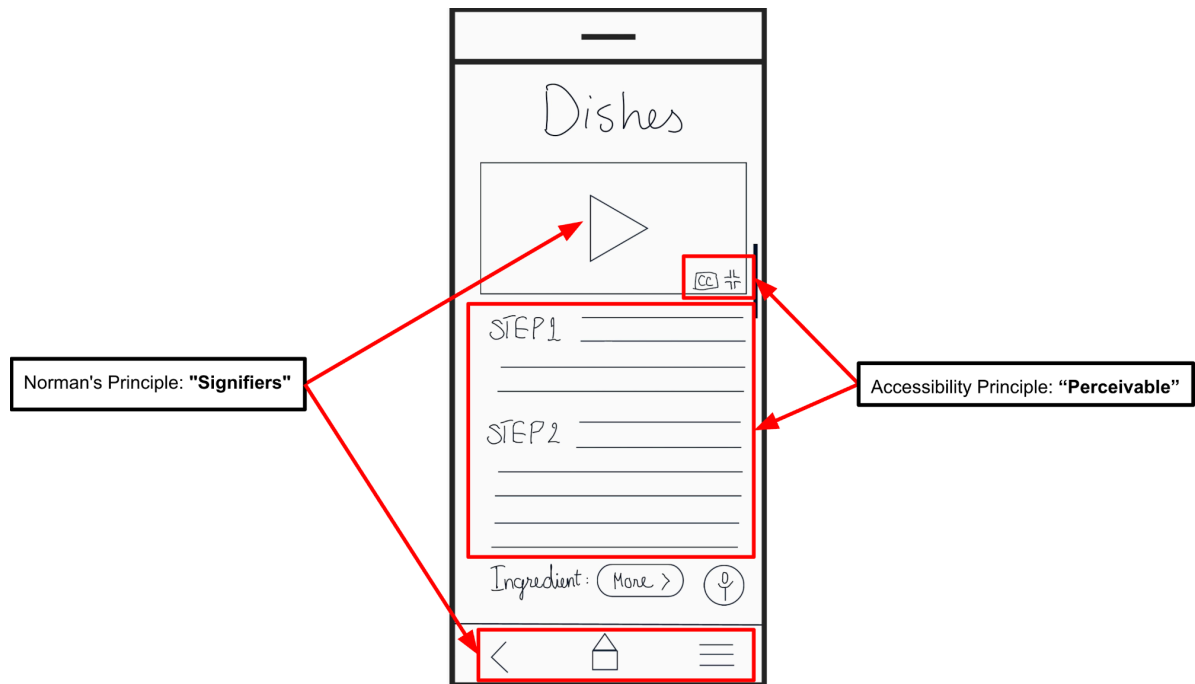


Photo 5: Second screen with annotation

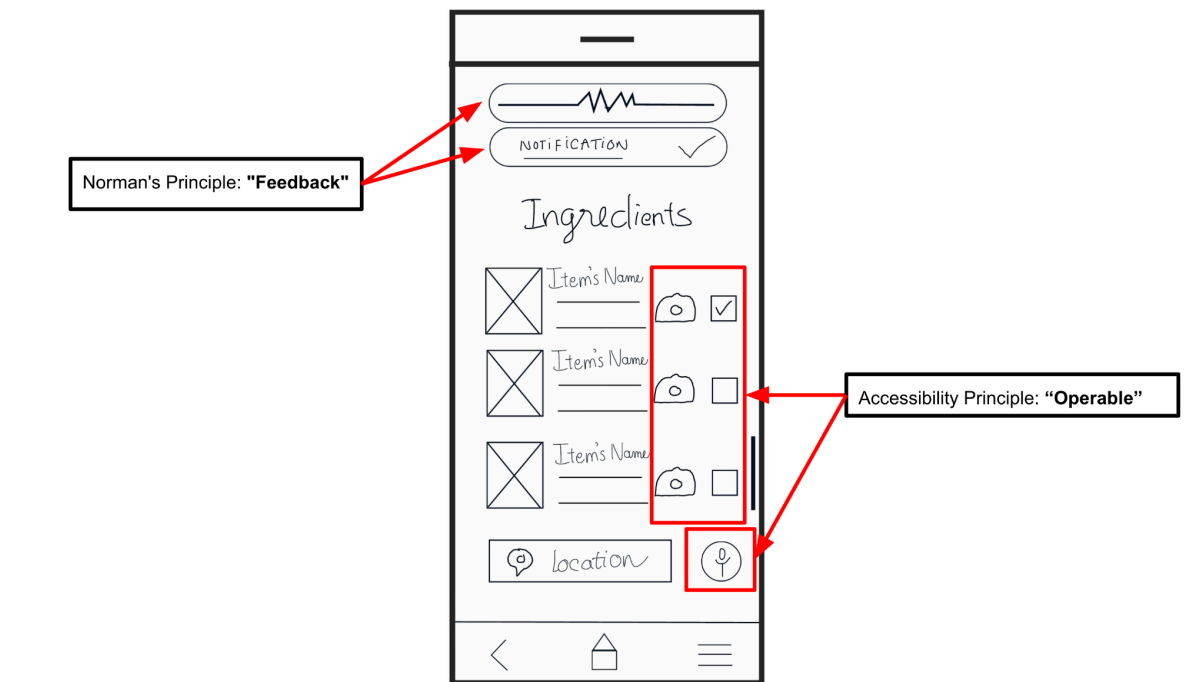


Photo 6: Third screen with annotation

6.Design Report

Screen code: LACY001

Address user stories: *As Lacy, I want to have a variety of cuisine and desserts to learn, so that I can have various meal options to cook for my family.*

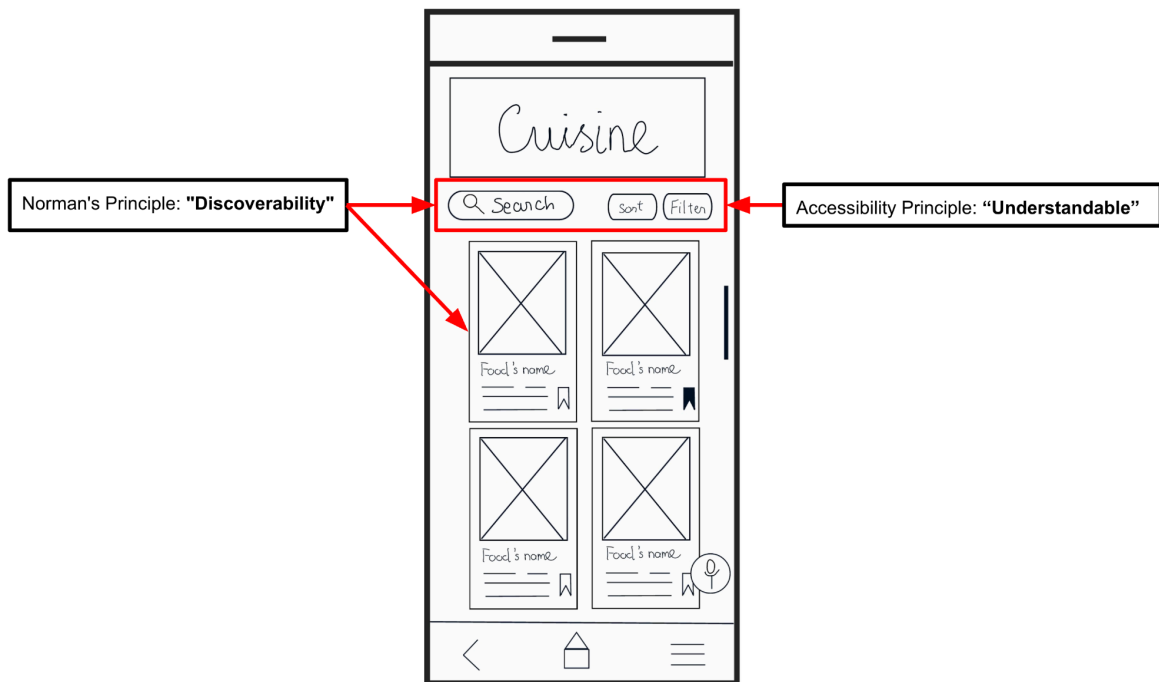


Photo 7: LACY001

Discoverability: All required functions in this section such as “search”, “sort”, “filter”, “save card” and “expand card details” for specific dishes are clearly displayed. Moreover, the “search”, “sort”, “filter” are positioned at the top of the page, which is a noticeable position. Each card contains its own “save” icon and “expand card details” (clickable card), helping users identify and take appropriate actions on their selected cards.

Understandable: The search, sort, filter action are clearly demonstrated and understandable to users. By using simple and well-known English words, users can identify and take action based on their needs. By looking at the “search” bar, users can predict that a typing action is required to find the food name. Clear labeling of the two buttons as “sort” and “filter” helps users avoid misuse and find dishes with less effort.

Screen code: LACY002

Address user stories:

- As Lacy, I want to have a variety of cuisine and desserts to learn, so that I can have various meal options to cook for my family.
- As Lacy, I want to know up-to-date details of the ingredients such as price and places to buy, so that I can find the appropriate ingredients with minimal effort.

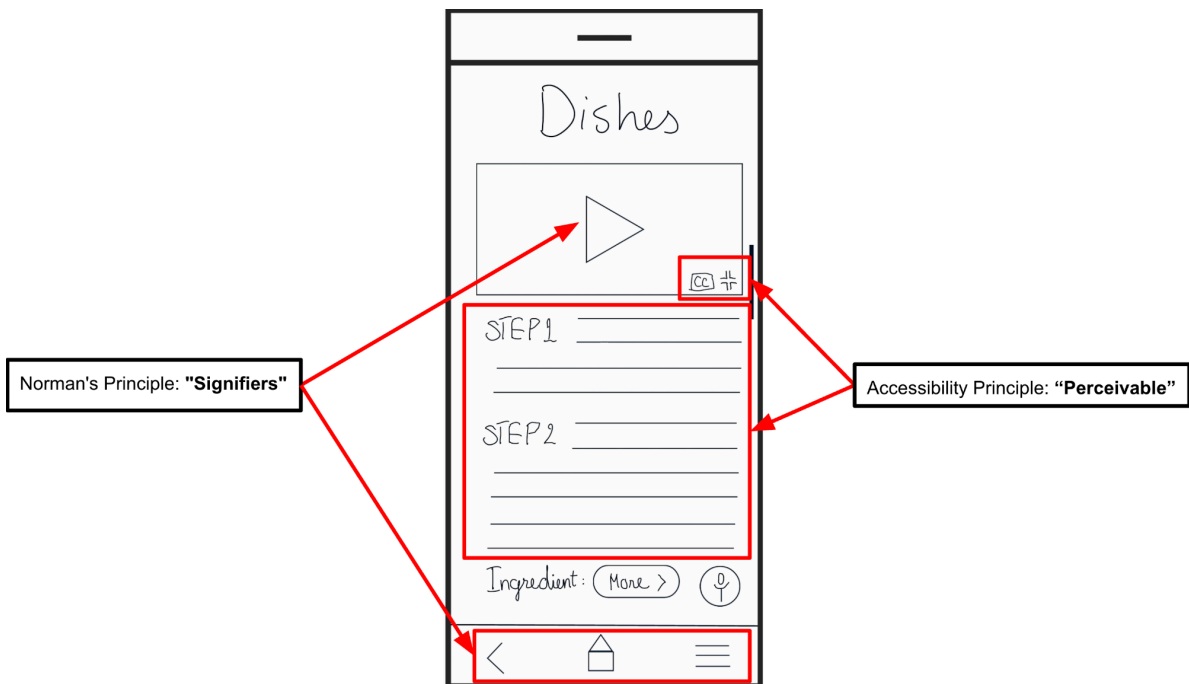


Photo 8: LACY002

Signifiers: Available actions can be understood through the physical shape of icons, providing simple instruction and giving users confidence to take appropriate actions. For example, a left-pointing arrow indicates moving back to the previous page, while a "home" icon signifies going to the home page, and the three horizontal stripes represent the menu. In addition, the "play" button demonstrates a forward motion, showing the start or continuation of a video.

Perceivable: While watching the video, users can find a "caption" button at the video corner, supporting those with hearing difficulties. An "expand video" button also allows users with vision impairment to change the video size for better visibility. Moreover, users can follow all steps through accompanying text paragraphs, which can also be read out loud by AI, ensuring accessibility to all users.

Screen code: LACY003

Address user stories: *As Lacy, I want to know up-to-date details of the ingredients such as price and places to buy, so that I can find the appropriate ingredients with minimal effort.*

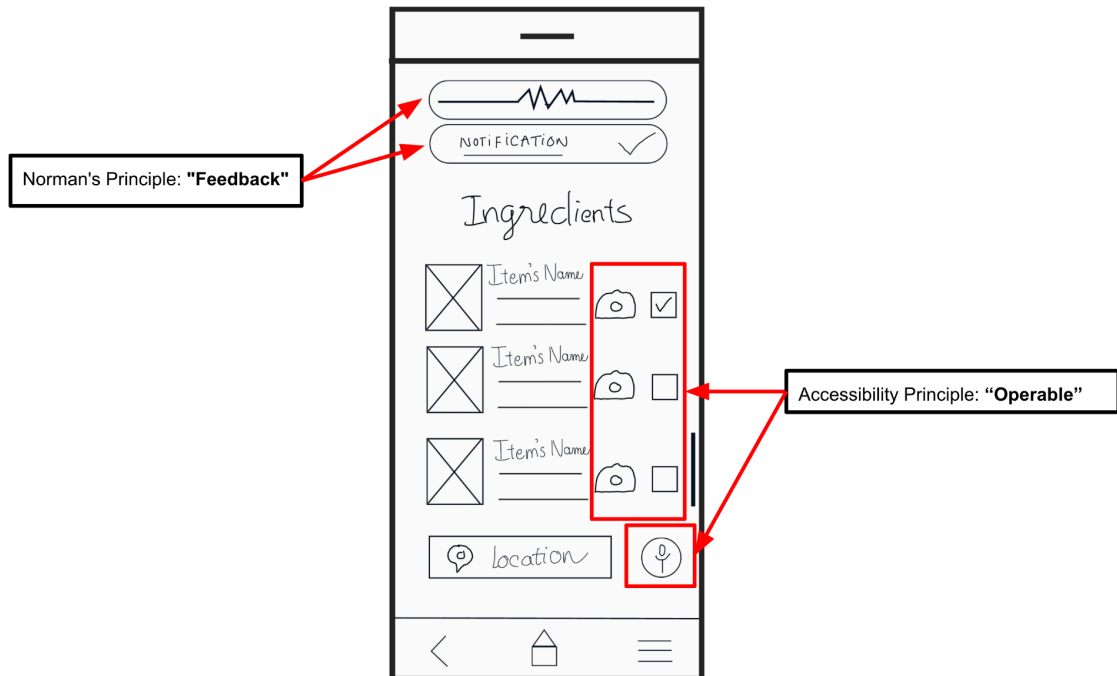


Photo 9: LACY003

Feedback: Whenever users activate the AI tool, the system will provide immediate feedback by displaying a “status” bar at the top of the screen once activation is complete. Users will be aware the action was successful and take further action. In addition, when the request of the user is completed, a notification will pop up to demonstrate the success action, providing a clear communication and user’s awareness.

Operable: Users with disabilities who can not manually tick checkboxes can activate AI tools by using voice commands or clicking the “microphone” button to tick all selected items. In addition, users with vision impairments can activate the camera scanning function through AI, allowing them to scan ingredients and automatically tick checkboxes. These additional features enable all users to complete tasks effectively, regardless of physical limitations.

Conclusion

Overall, the three low-fidelity prototypes above successfully meet all acceptance criteria and user requirements. They provide valuable ideas and visuals of the product's structure and functionality in the early stage. However, some limitations should be considered. Low-fidelity prototypes can not demonstrate the complete flow of the product and accurately present visual elements of the product such as colors, layout. In addition, they do not support testing and comparing features such as color contrast, text size, image size. For the upcoming high-fidelity prototype, The first important goal will be to comprehensively and accurately demonstrate the interface components such as color, text size, image. The second goal will aim to provide the product's interaction flow and the specific behavior of each component.

Appendix

Relevant Personas (from submission 1, based on selected user stories)



Generated Photos

" I WANT TO COOK VARIOUS DISHES FOR MY FAMILY "

LACY HO
SENIOR OFFICE WORKER

AGE 31
GENDER WOMAN
LOCATION MELBOURNE CBD
MARITAL STATUS MARRIED

BIOGRAPHY

Lacy is a senior at her company. She has to work overtime and can't cook everyday. Her family just moved to a new apartment in Melbourne CBD near her company, which can be a challenge to her family finances. Now, she wants to spend about 2 hours or more per week on cooking and learning new cuisine such as Asian, Western and Desserts to prevent eating out, improve her cooking skills, health and her family meals' quality.

MOTIVATION

- Family
- Finance
- Foods

GOALS

- Improving personal health and cooking skills.
- Want to learn how to cook Desserts, Asian and Western cuisines.
- Understanding and cooking any recipes with less effort and time.
- Cook good and diversified foods for her kids and husband, improving the family meals.

PERSONALITY

Detailed focus ●●●●●

Problem-Solving ●●●●●

Concentration ●●●●●

Social media ●●●●●

SKILLS

Cooking skills

Technology

Learning

FRUSTRATIONS

- Facing health issues such as low vision and Lactose Intolerance.
- Lack of time in cooking because of the nature of the job.
- Suffering from affording and finding ingredients.
- Lack of cooking experience, suffering from following complex cooking instructions.

[1]: Image generated using Generated Photo: <https://generated.photos/face/neutral-asian-young-adult-female-with-long-brown-hair-and-brown-eyes> , with the following specifications: Neutral asian young-adult female with long brown hair and brown eyes.